
ITCS3166

Computer Networks

Lecture 1 Introduction

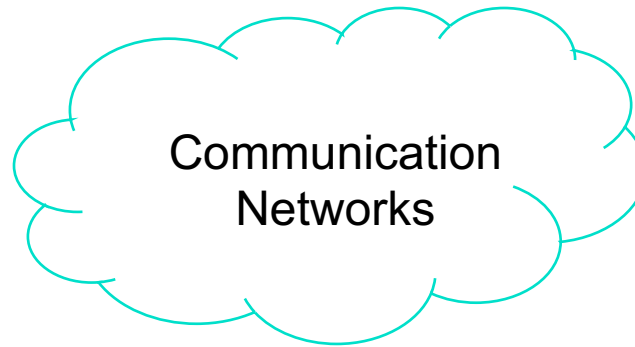
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Why learn about the Network and Internet?

1. Curiosity »
2. Impact on our world »
3. Job prospects!



What are communication networks?

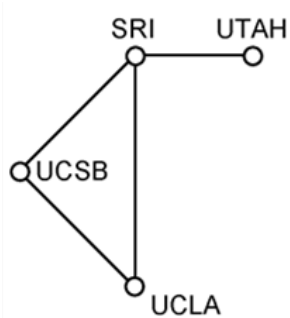


- **The equipment (hardware & software) and facilities that provide the basic communication service**
 - **A computer network is a collection of autonomous computers interconnected by a single technology**
- **Equipment**
 - **Routers, servers, switches, multiplexers, hubs, modems, ...**
- **Facilities**
 - **Copper wires, coaxial cables, optical fiber**
 - **Ducts, conduits, telephone poles ...**

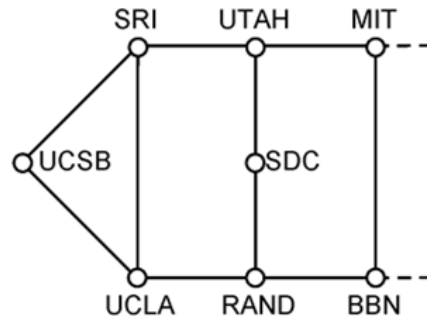
How are communication networks designed and operated?

From this experimental network ...

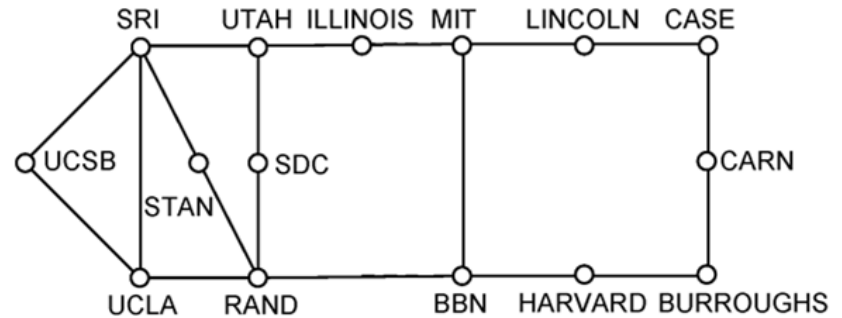
ARPANET ~1970



(a) Dec. 1969.



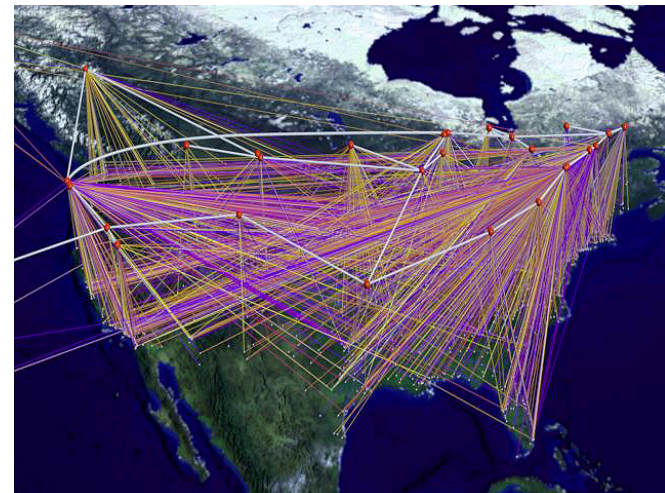
(b) July 1970.



(c) March 1971.

Today's Internet

- An everyday institution used at work, home, and on-the-go
- Visualization contains millions of links



Uses of Computer Networks

- **Business Applications**
- **Home Applications**
- **Mobile Users**
- **Social Issues**

What applications and services you are familiar?

Internet – Societal Impact

- **An enabler of societal change**
 - **Easy access to knowledge**
 - **Electronic commerce**
 - **Personal relationships**
 - **Discussion without censorship**



WIKIPEDIA

PayPal

match.com 

Tor 

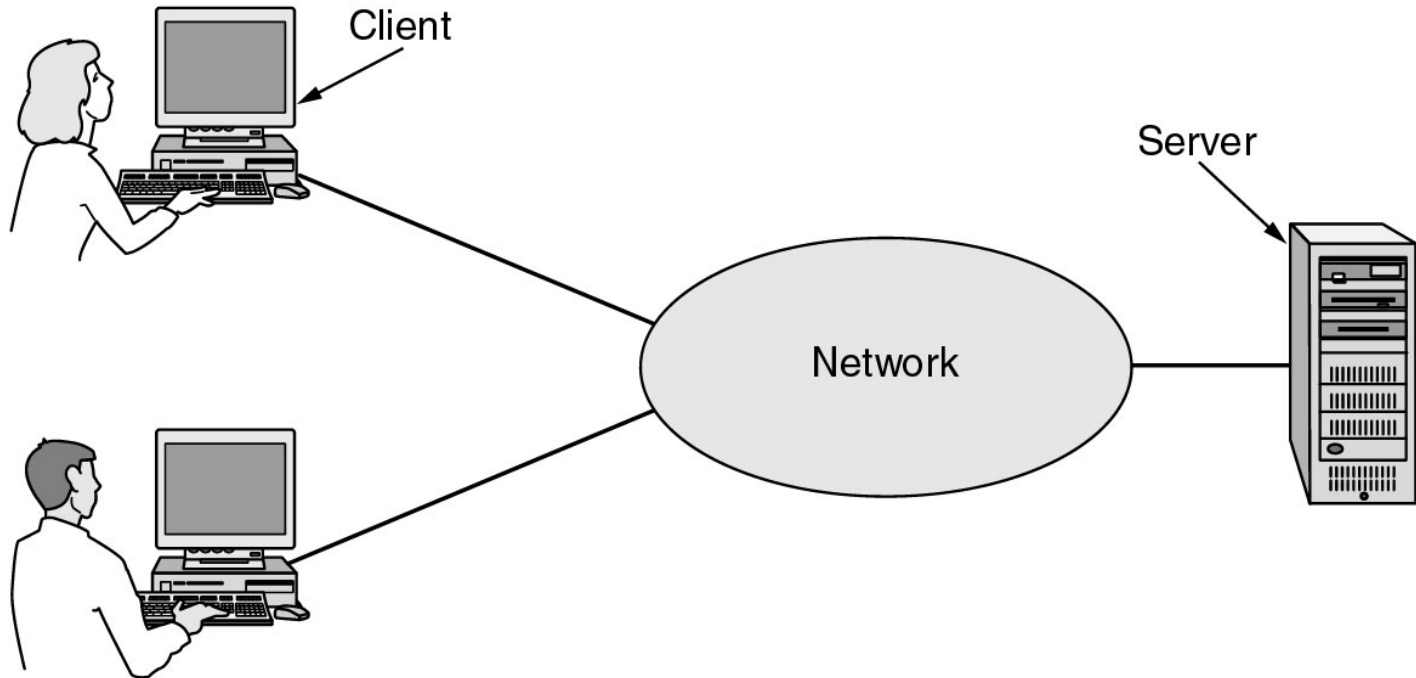
Internet – Economic impact

- ° An engine of economic growth
 - Advertising-sponsored search
 - “Long tail” online stores
 - Online marketplaces
 - Crowdsourcing



Client/Server Model - Functionality

- **Business Applications of Networks: a network with two clients and one server.**

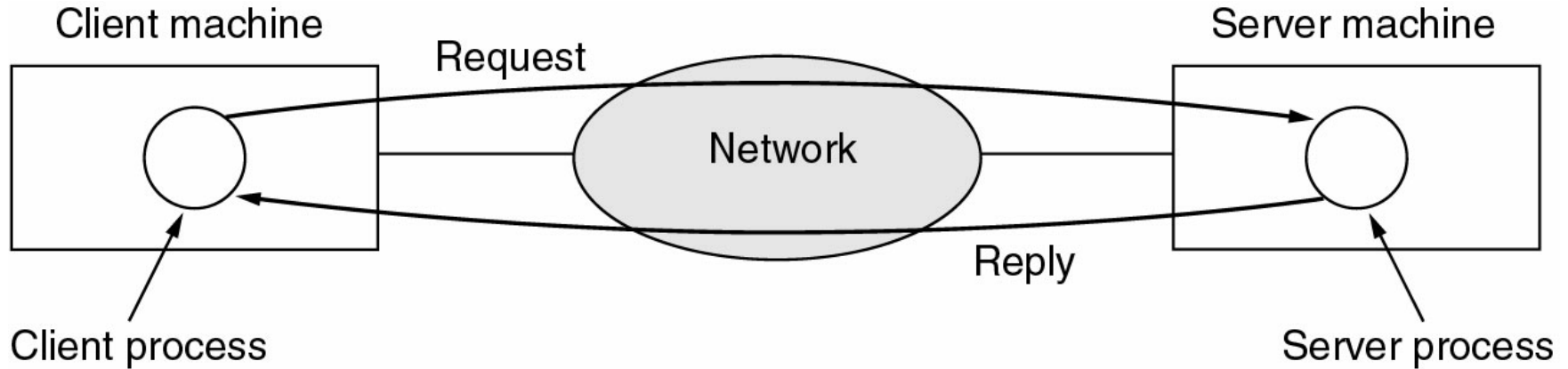


What are the key issues here?

How clients and the server communicate and interact?

Client/Server Model(2) - Reliability

- The client-server model involves **requests** and **replies**.



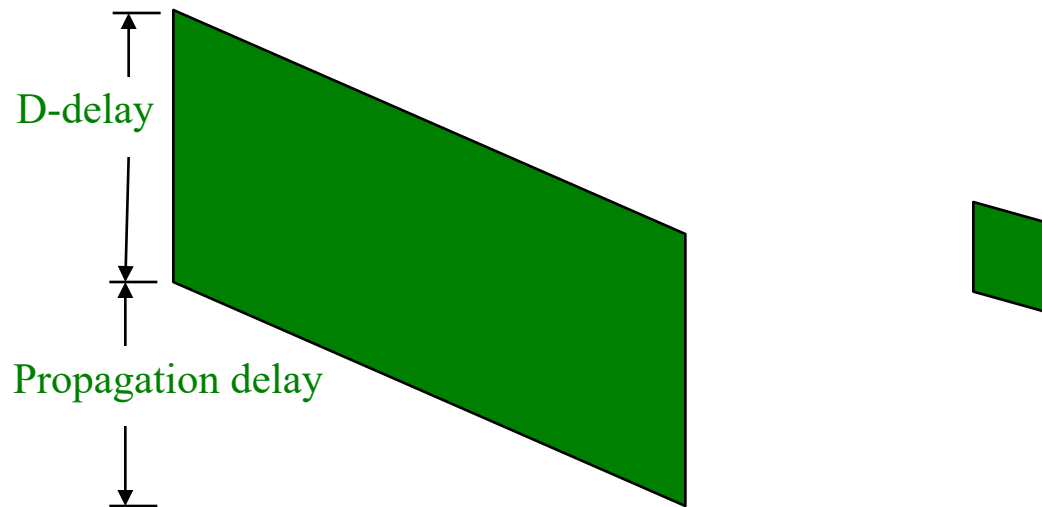
What happens if the reply got lost?

Do you like that your money saving request got done twice?

Client/Server Model(3) - Performance

- The performance of a client-server system is influenced by two factors: the **bandwidth** of the network (how many bits/sec it can transport) and the **propagation latency** (how much time it takes for the first bit to get from the client to the server).

Please give an example of a network that exhibits high bandwidth and high latency. Then give an example of one with low bandwidth and low propagation latency.



Propagation delay is independent to the bandwidth, but the distance.

Home Network Applications

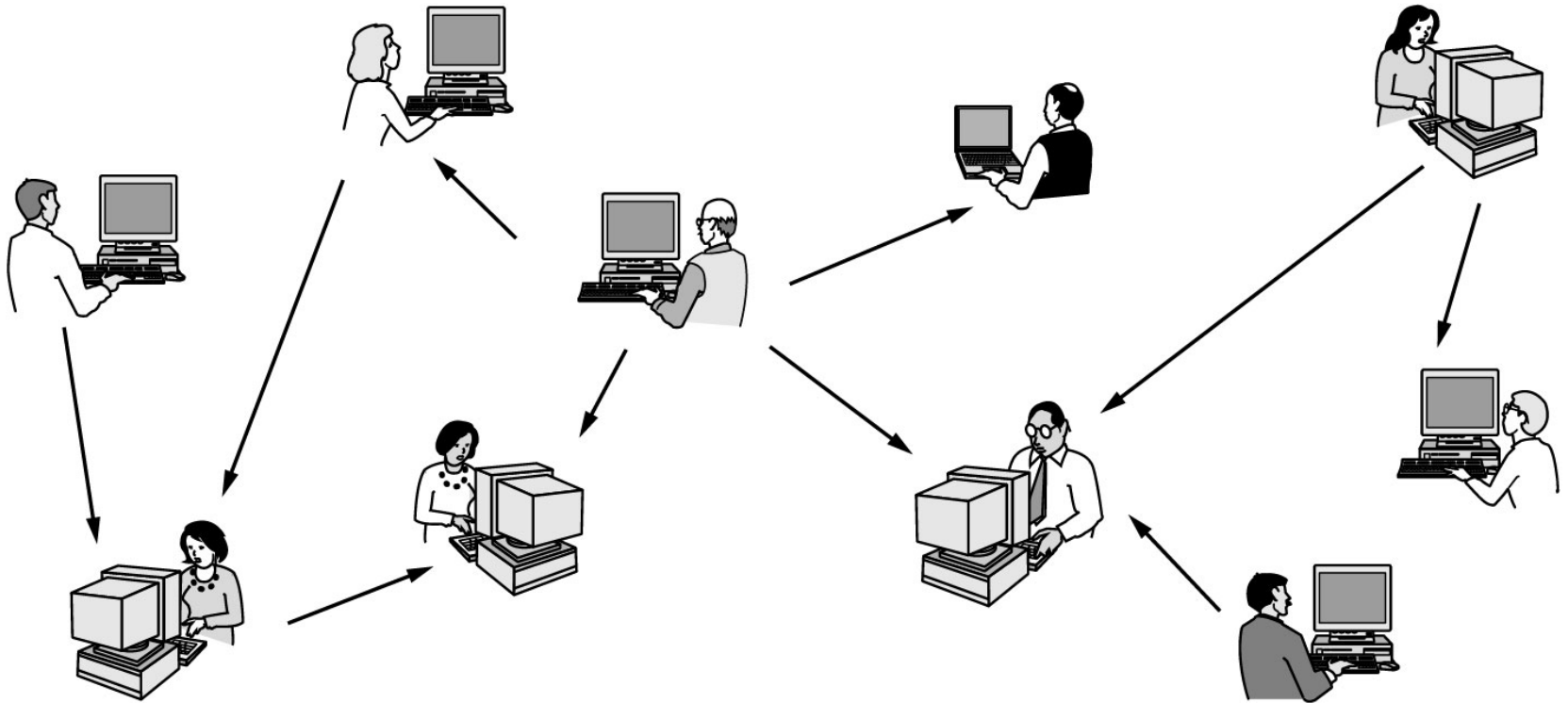
- **Access to remote information**
- **Person-to-person communication**
- **Interactive entertainment**
- **Electronic commerce**

Examples?

What are important things in those applications?

Home Network Applications (2)

- In peer-to-peer system there are no fixed clients and servers.



Why Napster was shut down?

Services & Applications

- **Service: Basic information transfer capability**
 - Internet transfer of individual block of information
 - Internet reliable transfer of a stream of bytes
 - Real-time transfer of a voice signal
- **Applications build on communication services**
 - E-mail & web build on reliable stream service
 - Fax and modems build on basic telephone service
- **New applications build on multiple networks**
 - SMS (short message service) builds on Internet reliable stream service and cellular telephone text messaging

Mobile Network Users

- **Combinations of wireless networks and mobile computing.**

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory

Network Hardware

- **Local Area Networks**
- **Metropolitan Area Networks**
- **Wide Area Networks**
- **Wireless Networks**
- **Home Networks**
- **Internetworks**

Classifications: Size, transmission technology, and topology

Transmission Technology Classification

- **Two Types of transmission technology**
 - **Broadcast links**
 - **Point-to-point links**
- **Broadcast Networks have a single communication channel that is shared by all the machines on the network**
 - **Can a broadcast network support 1-1, 1-many communications? And How?**
- **Point-to-point networks consist of many connections between individual pairs of machines**
 - **Can a point-to-point network support 1-many and 1-all communications? And how?**

General rule: smaller and geographically localized networks tend to use broadcasting while larger networks usually are point-to-point.

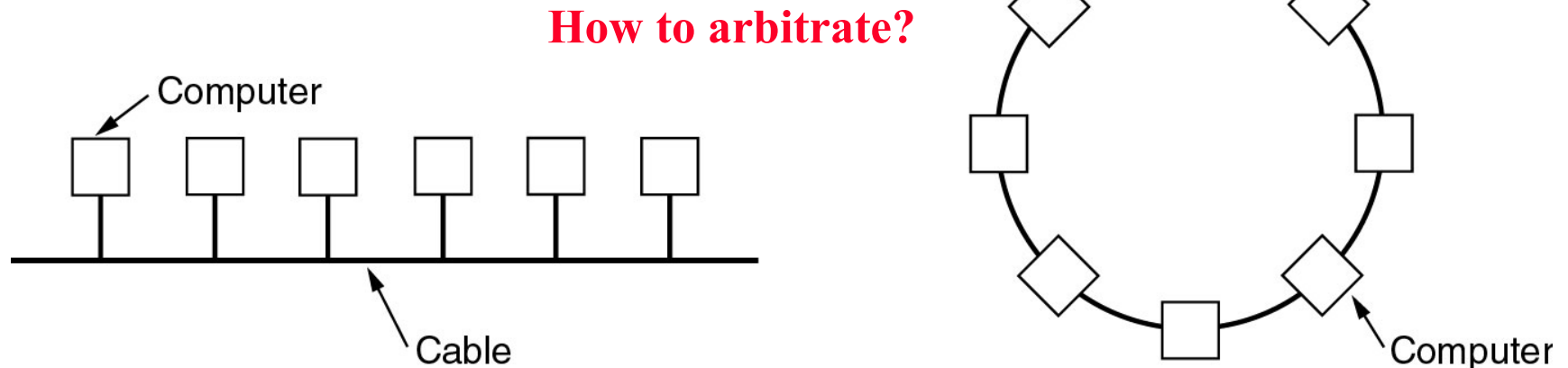
Scale Classification

- **Classification of networks by scale.**

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Local Area Networks (LAN)

- LANs are privately-owned networks within a single building or campus of up to a few KMs in size.
- How LANs are distinguished from other networks?
 - Size
 - Transmission technology
 - topology



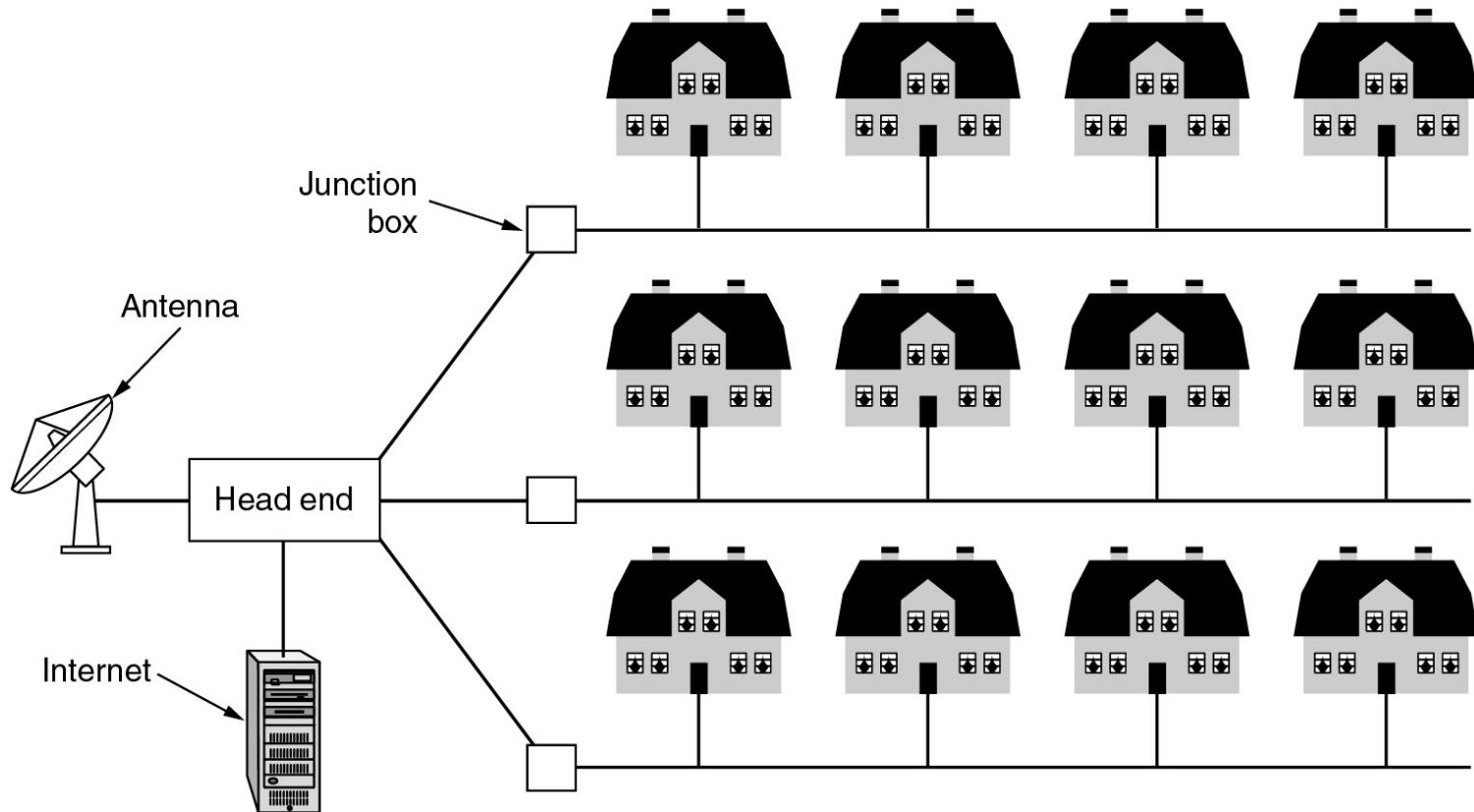
Two broadcast networks (a) Bus-based, (b) Ring

What is the benefit from the restricted network size?

What happens if more than one machine want to transmit?

Metropolitan Area Networks (MAN)

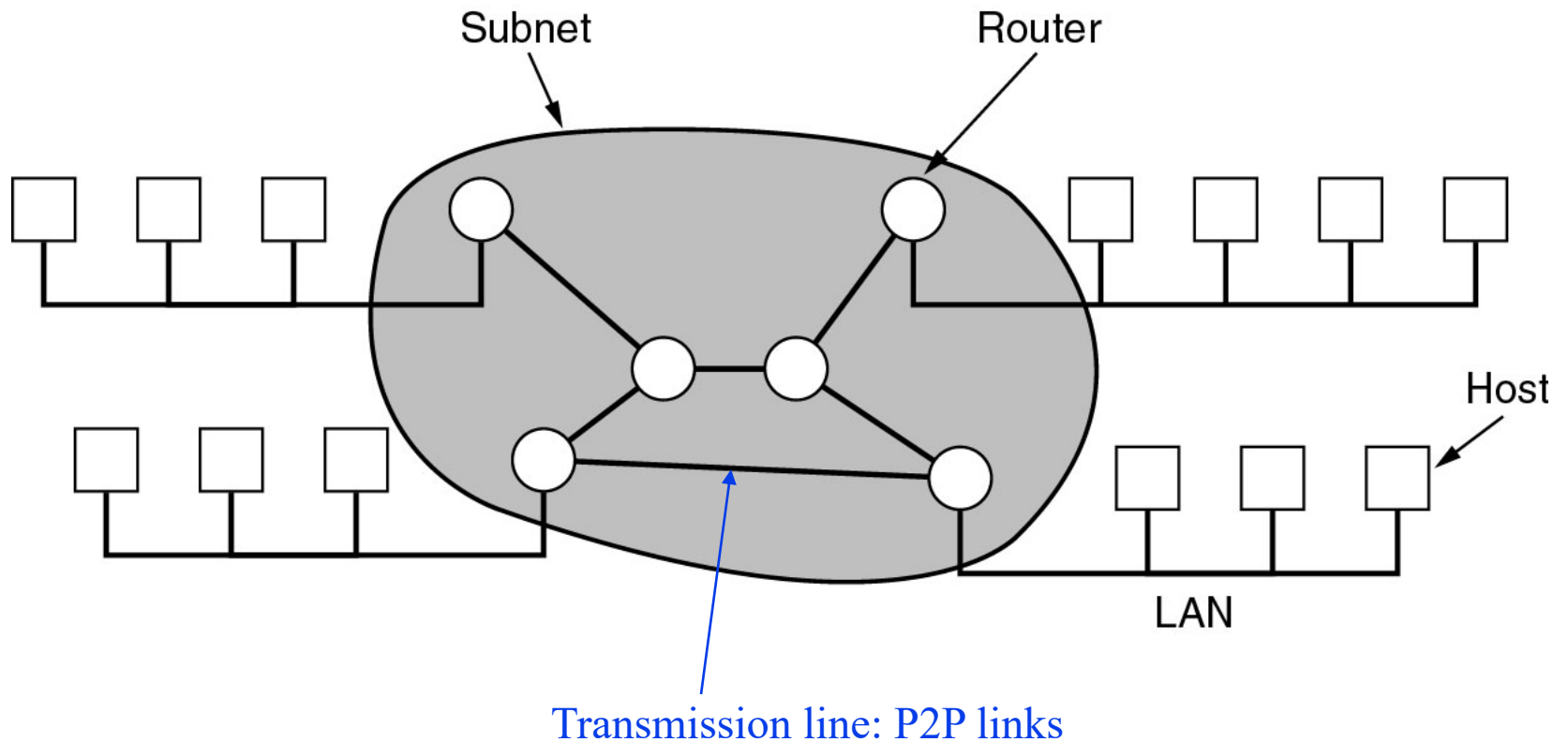
- A metropolitan area network based on cable TV.



Is the cable always faster than DSL?

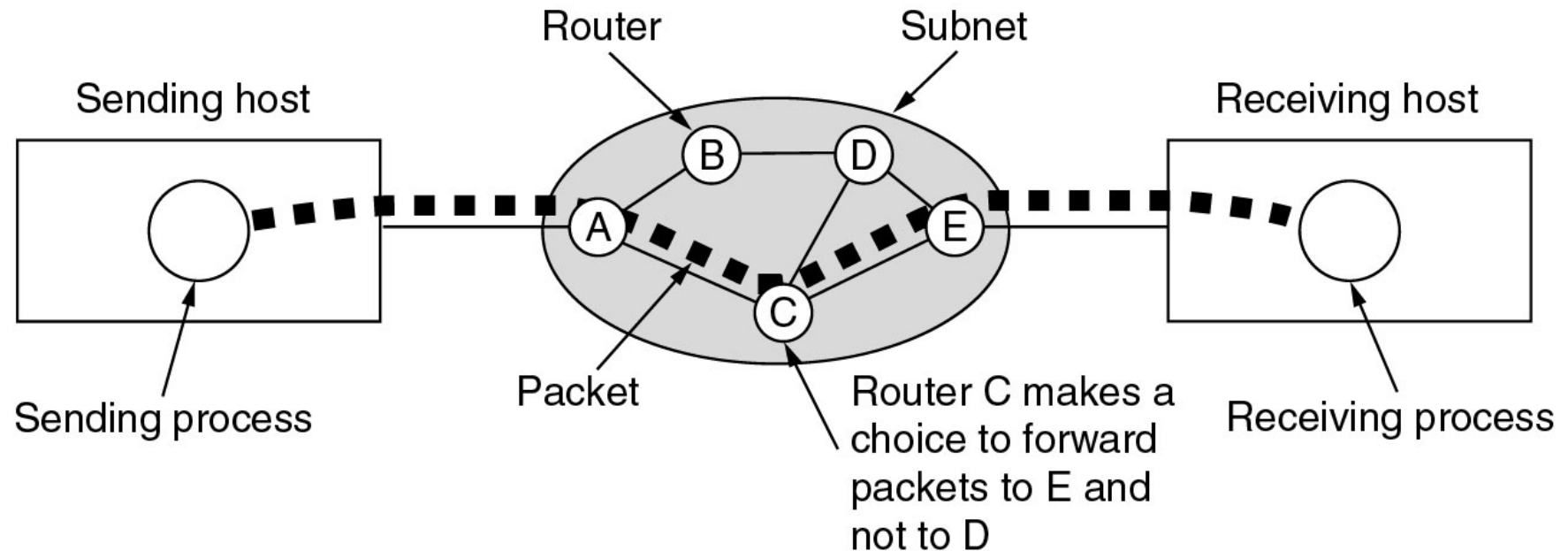
Wide Area Networks (WAN)

- **Relation between hosts on LANs and the subnet.**



Wide Area Networks (2)

- **Store-and-forward (packet-switched):** a packet is stored in an intermediate router **in its entirety**, stored there until the required output line is free, and then forwarded.



A stream of packets from sender to receiver.

Must all packets of a message follow the same route?

How packets are forwarded at the router?

Wide Area Networks (3)

- A collection of **4** routers is to be connected by a point-to-point subnet. Between each pair of routers, the designers may put a high-speed line and a low-speed line. If it takes **100ms** of computer time to generate and inspect each topology, how long will it take to inspect all of them?

64*100ms

Wide Area Networks (4)

- A factor in the delay of a store-and-forward packet-switching system is how long it takes to store and forward a packet through a network. If switching is $10\ \mu\text{sec}$, is this likely to be a major factor in the response of a client-server system where the client is in NYC and the server is in LA? Assume the propagation speed in copper and fiber to be $2/3$ of the speed of light in vacuum.

No. The speed of propagation is 200,000 km/sec or 200 meters/microsec. In 10 microsec the signal travels 2km. Thus, each switch adds a equivalent of 2kms of extra cable. If the client and server are seperated by 5000 km, traversing even 50 switches adds only 100km to the total path, which is only 2%. Thus, switching delay is not a major factor under these circumstances.

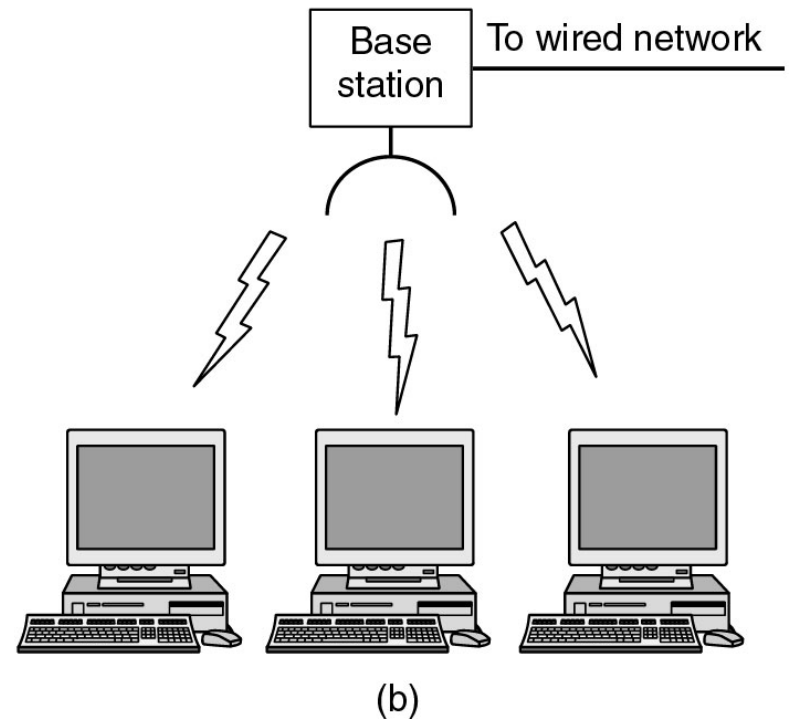
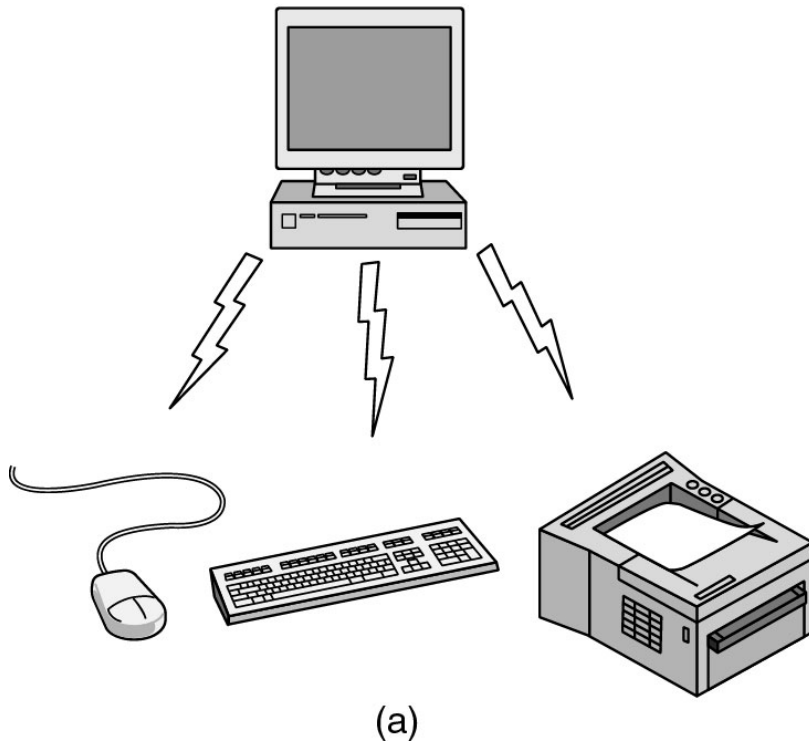
Wireless Networks

Three Categories of wireless networks:

- **System interconnection**
- **Wireless LANs**
- **Wireless WANs**

Wireless Networks (2)

- (a) Bluetooth configuration for system interconnection
- (b) Wireless LAN (802.11)



Why master-slave architecture in bluetooth?

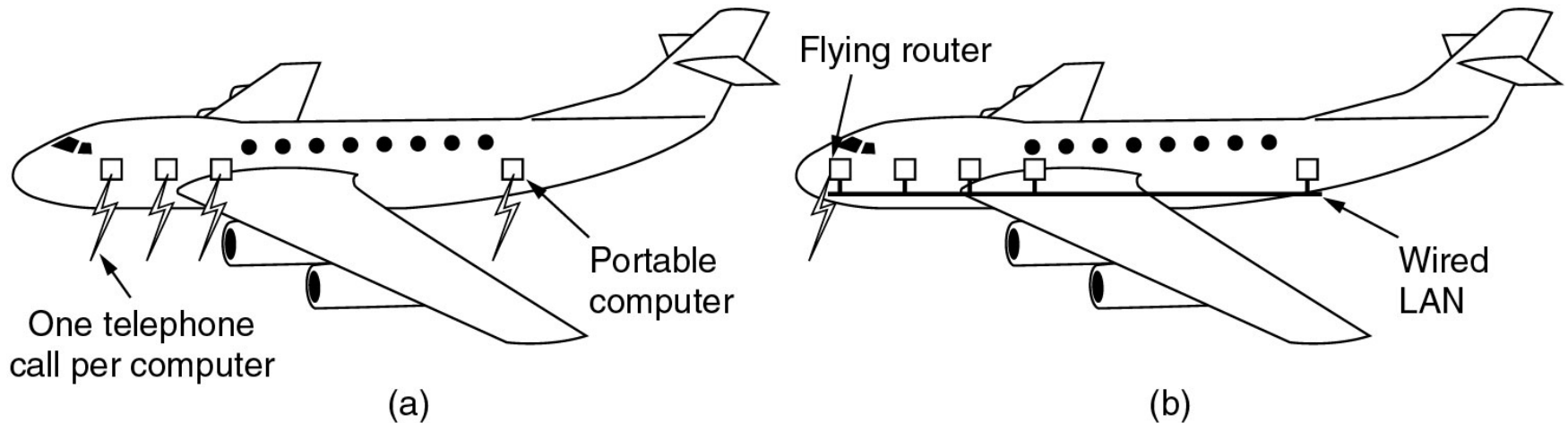
Wireless Networks (3)

- **Wireless WANs**
 - **The radio network for cell phones (3G)**
 - **The local multipoint distribution service (802.16)**

Bandwidth vs. scope/coverage

Wireless and Wired Networks

- Almost all wireless networks hook up to the wired network at some point to provide access to the Internet.



(a) Individual mobile computers **(b) A flying LAN**

Home Network Categories

Devices are capable of being networked:

- **Computers (desktop PC, PDA, shared peripherals)**
- **Entertainment (TV, DVD, VCR, camera, stereo, MP3)**
- **Telecomm (telephone, cell phone, intercom, fax)**
- **Appliances (microwave, fridge, clock, furnace)**
- **Telemetry (utility meter, burglar alarm, babycam).**

What they need to be networked?

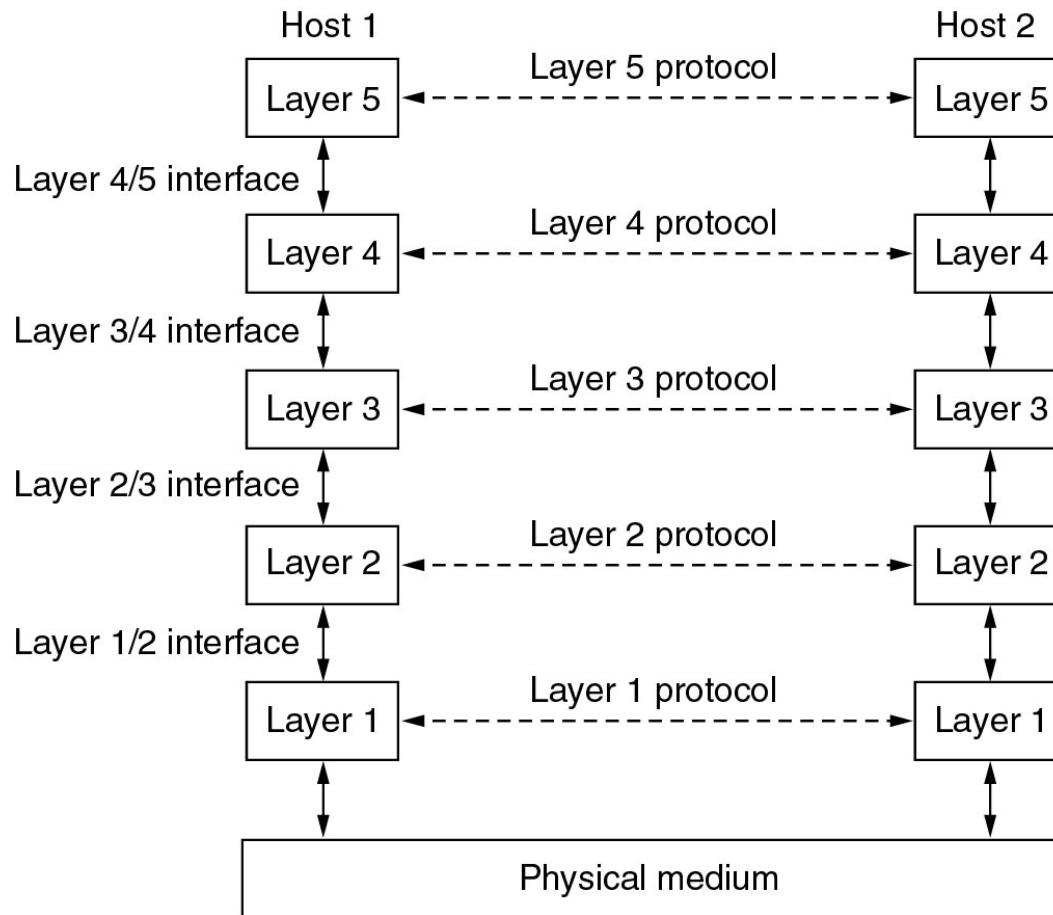
Embedded computing + mobile computing → ubiquitous computing

Network Software

- **Protocol Hierarchies**
- **Design Issues for the Layers**
- **Connection-Oriented and Connectionless Services**
- **Service Primitives**
- **The Relationship of Services to Protocols**

Network Software: Protocol Hierarchies

- **Networks are organized as a stack of layers/levels, each one built upon one below it.**

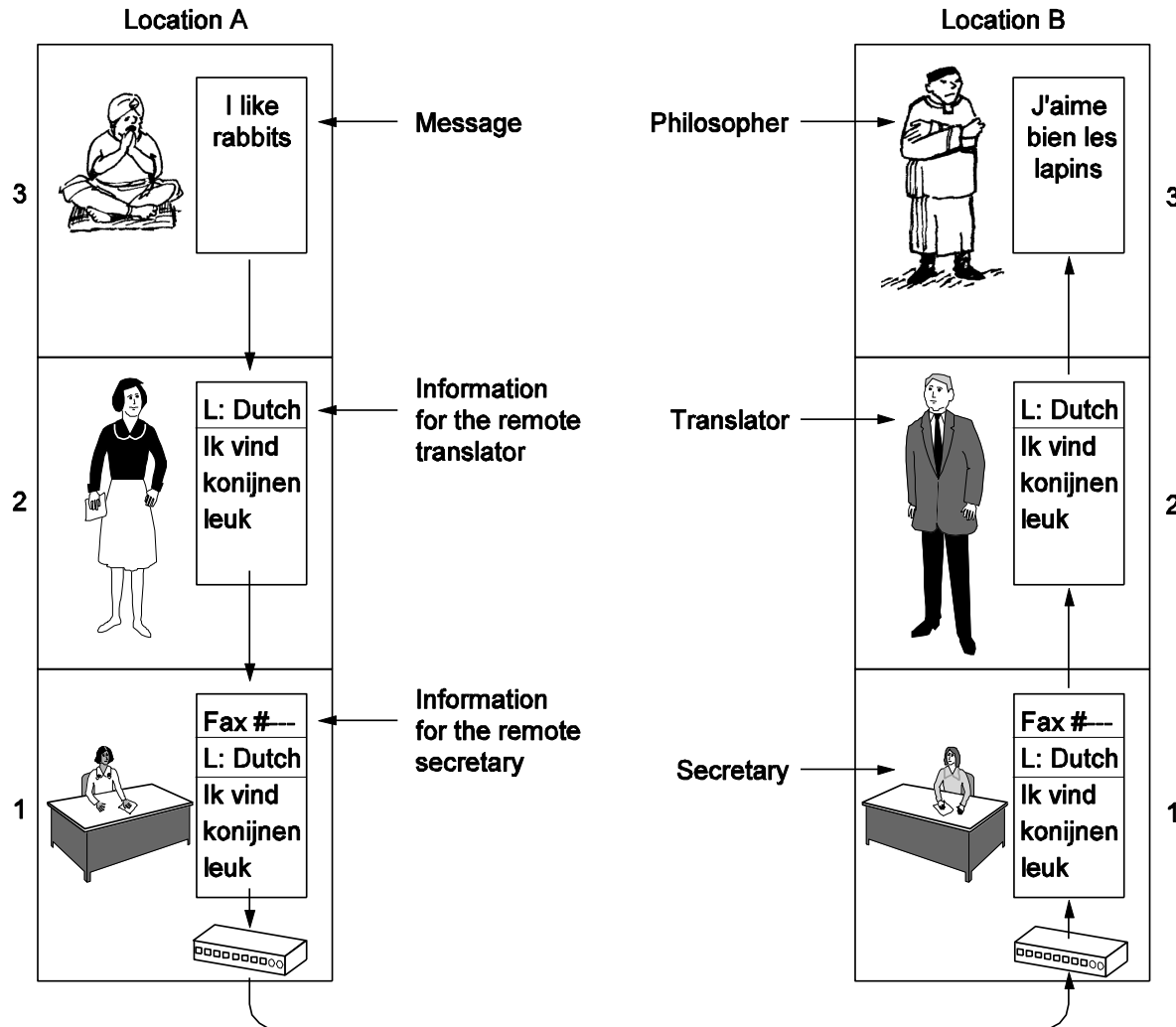


**Why hierarchical
Network architecture?**

Layers, protocols, and interfaces.

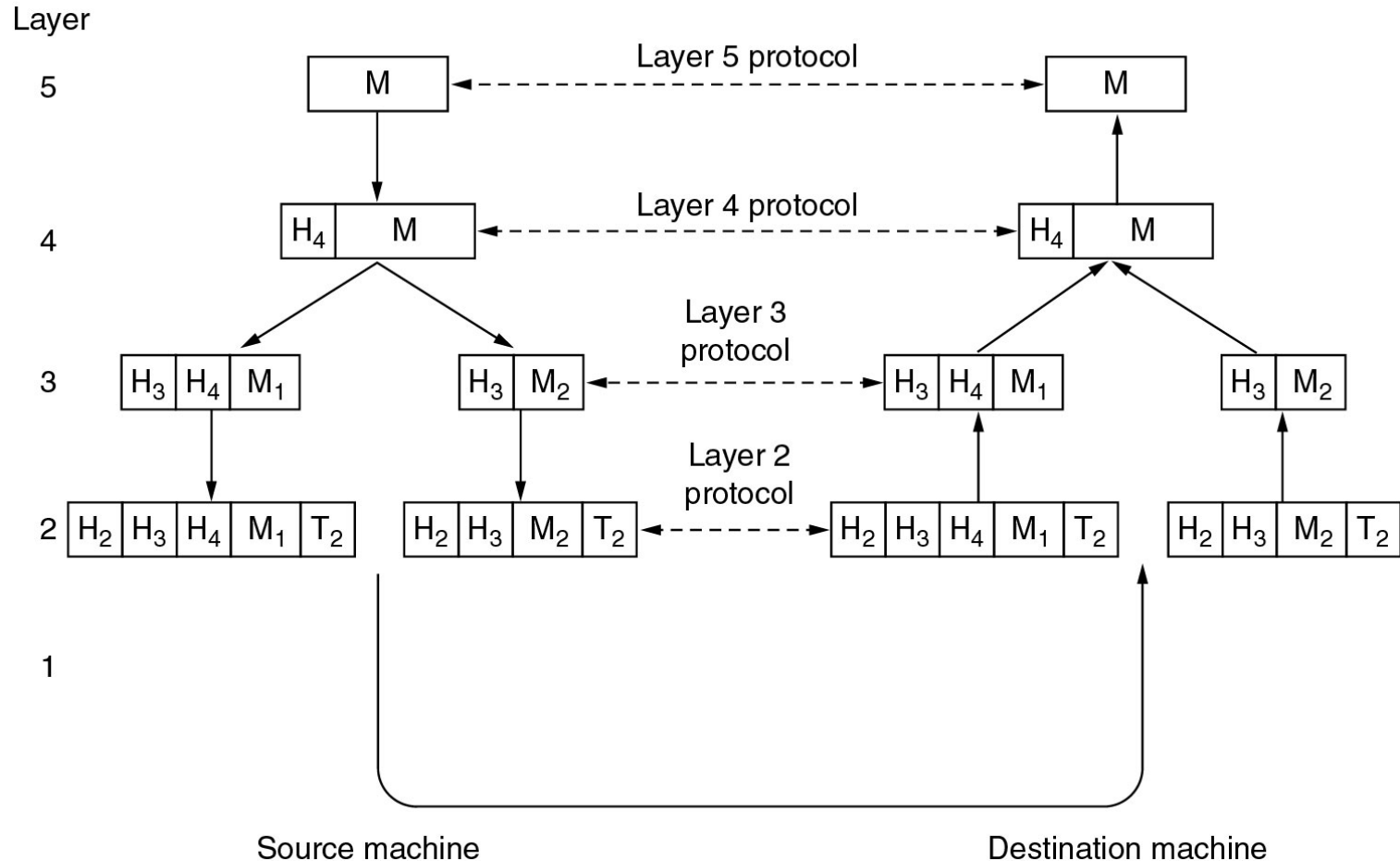
Protocol Hierarchies (2)

- The philosopher-translator-secretary architecture.



Protocol Hierarchies (3)

- The relation between the virtual and actual communication and the difference between protocols and interfaces.



What is the effective bandwidth utilization?

Design Issues for the Layers

- **Addressing**
- **Framing**
- **Error Control**
- **Flow Control**
- **Ordering**
- **Multiplexing**
- **Routing**
- **Congestion control**
- **Quality of Service (QoS)**

Connection-Oriented and Connectionless Services

- Layers can offer two different type of services to the layer above them: telephone systems vs. postal systems.

		Service	Example
Connection-oriented	{	Reliable <u>message stream</u>	Sequence of pages
		Reliable <u>byte stream</u>	Remote login
		Unreliable connection	Digitized voice
Connection-less	{	Unreliable datagram	Electronic junk mail
		Acknowledged datagram	Registered mail
		Request-reply	Database query

Six different types of service.

What is reliable and unreliable communication? Think about postal services

Can a unreliable service be made reliable by using a network software?

Why would anyone actually prefer unreliable to reliable communication?

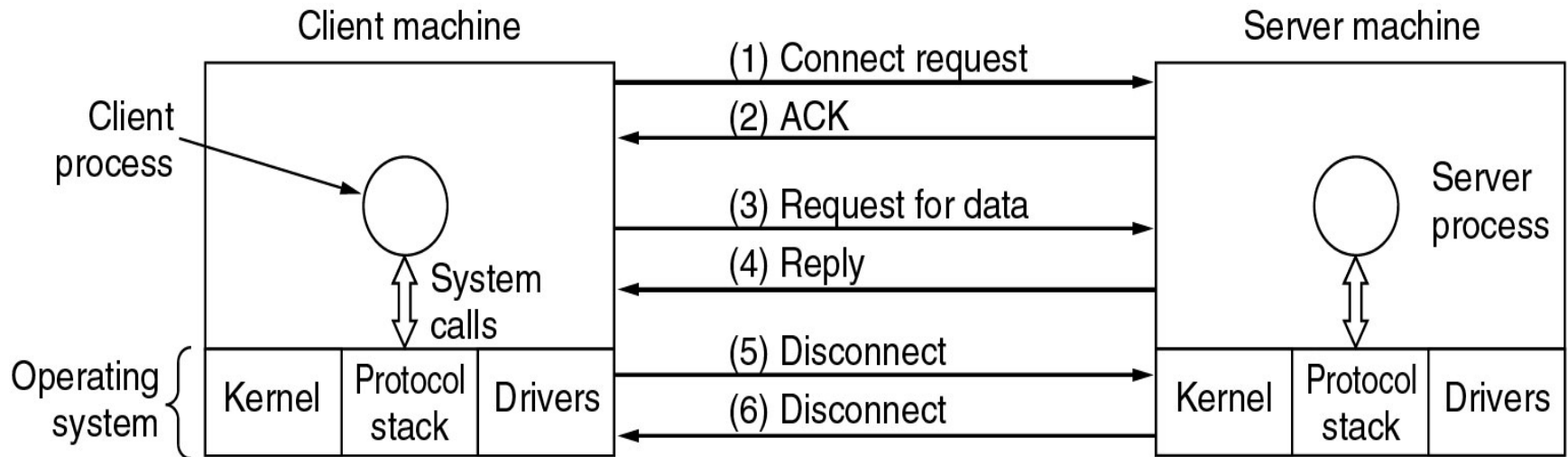
Service Primitives

- A **service** is specified by a set of **primitives** (operations) available to a user process to access the service.

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Five service primitives for implementing a simple connection-oriented service.

Service Primitives (2)



Packets sent in a simple client-server interaction on a connection-oriented network.

Life is not so simple. What could be wrong?

Why a connectionless protocol is not used instead?

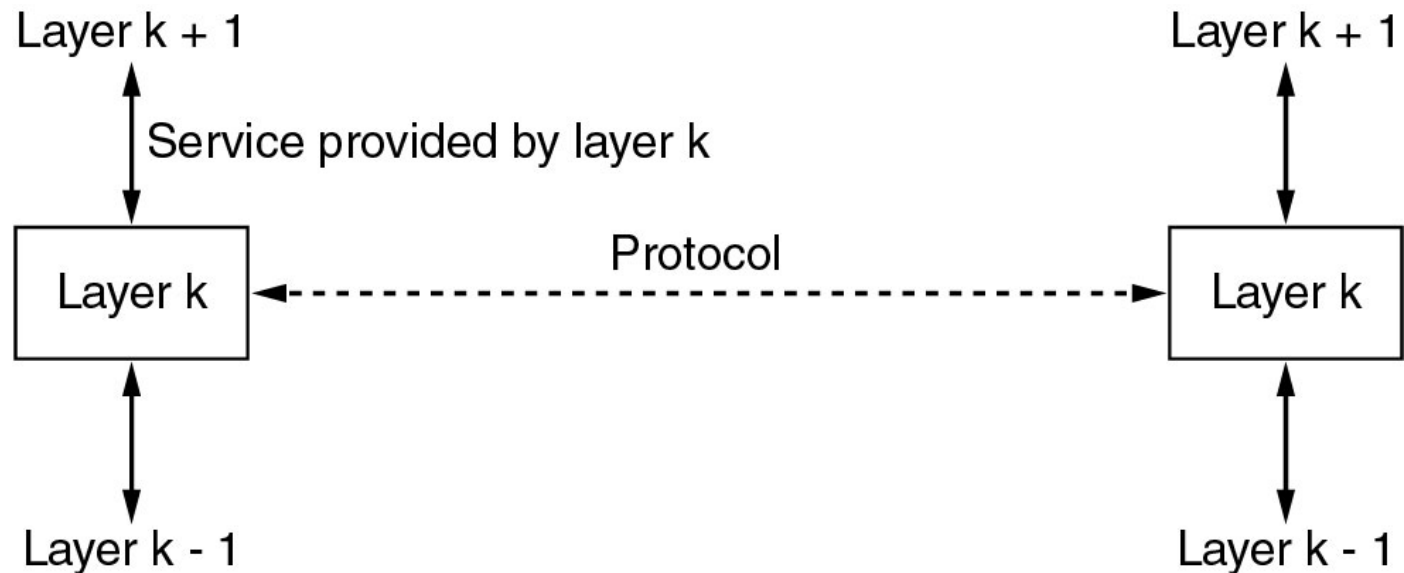
ACK Strategies

- **To transfer a file between two computers, two acknowledgement strategies are possible.**
- **In the first one, the file is chopped up into packets, which are individually acknowledged by the receiver.**
- **In the second one, the packets are not acknowledged individually, but the entire file is acknowledged when it arrives.**

What are advantages and disadvantage of these two strategies?

Services to Protocols Relationship

- A **service** is a set of **primitives** (operations) available to a user process to access the service
- A **protocol** is a set of **rules/agreements** governing the format and meaning of the packets, or messages that are exchanged by the peer entities within a layer (relates to the implementation of a service)



The relationship between a service and a protocol.

Are there any other services implicitly in this figure?

Questions?

- **Suppose the algorithms used to implement the operations at layer k is changed. How does this impact operations at layers $k - 1$ and $k + 1$?**
- **Suppose there is a change in the service (set of operations) provided by layer k . How does this impact services at layers $k-1$ and $k+1$?**

Reading

- **Chapter 1 of the textbook.**